

Numerical Analysis (SCI19 3111) — Final Cheat Sheet

Regression · Integration · ODEs · Linear Algebra (+ midterm recap)

1 — Regression / Data Fitting (Th.1–6)

Regression $\neq$ interpolation: interpolation trusts every measurement and passes through ALL points; regression assumes noise and finds the best OVERALL trend, passing through (almost) none. Deviations $d_i = y_i - \phi(x_i)$, data (x_i, y_i) , $i = 0..n$ ($n+1$ points). **Error choices:** naive $S = \sum d_i$ (rejected: $+$, $-$ cancel); abs $S = \sum |d_i|$ (rejected: kink at 0, no derivative); **squared** $S = \sum d_i^2$ (chosen: smooth everywhere, so “set gradient = 0” gives solvable linear equations; big misses punished quadratically). Example (Lec14): pts $(1, 2), (2, 1)$, line through $(1, 1), (2, 2)$: $d_0 = 1, d_1 = -1$: naive $S = 0$ (cancels!), abs $S = 2$, squared $S = 2$.

Linear regression

$\hat{y} = a_0 + a_1x$, $S = \sum (y_i - a_0 - a_1x_i)^2$. General fit $\phi(x, a_0..a_\ell)$: $\frac{\partial S}{\partial a_k} = -2 \sum_{i=0}^n (y_i - \phi(\dots)) \frac{\partial \phi}{\partial a_k} = 0$, $k = 0..\ell$ ($\ell+1$ equations). For the line this gives the two normal equations below. ∂_{a_0} : $\sum y_i - (n+1)a_0 - a_1 \sum x_i = 0$; ∂_{a_1} : $\sum x_i y_i - a_0 \sum x_i - a_1 \sum x_i^2 = 0$. Divide by $n+1 \Rightarrow a_0 + a_1 \bar{x} = \bar{y}$, $a_0 \bar{x} + a_1 \bar{x}^2 = \bar{x}\bar{y}$; Cramer gives

$$a_1 = \frac{\bar{x}\bar{y} - \bar{x}\bar{x}}{\bar{x}^2 - \bar{x} \cdot \bar{x}}, \quad a_0 = \bar{y} - a_1 \bar{x}$$

with $\bar{x} = \frac{1}{n+1} \sum x_i$, $\bar{x}^2 = \frac{1}{n+1} \sum x_i^2$, $\bar{x}\bar{y} = \frac{1}{n+1} \sum x_i y_i$.

Trap: $\bar{x}^2 \neq \bar{x}^2$; denominator = 0 iff all x_i equal (no fit possible).

Hessian + minimum test

$$H = \begin{pmatrix} S_{a_0 a_0} & S_{a_0 a_1} \\ S_{a_1 a_0} & S_{a_1 a_1} \end{pmatrix}$$

$$S_{a_0 a_0} = 2(n+1), \quad S_{a_0 a_1} = 2 \sum x_i, \quad S_{a_1 a_1} = 2 \sum x_i^2$$

$$\det H = 4 \left[(n+1) \sum x_i^2 - \left(\sum x_i \right)^2 \right]; \min \iff S_{a_0 a_0} > 0 \text{ AND } \det H > 0$$

Intuition: H = error-bowl curvature;

Cauchy–Schwarz $\Rightarrow \det H \geq 0$, > 0 if ≥ 2 distinct $x_i \Rightarrow$ unique global min.

Multivariable + pseudoinverse

$\hat{y} = a_0 + a_1x_1 + \dots + a_mx_m$, X is $(n+1) \times (m+1)$ with first col of 1's. $S = \|\hat{y} - X\hat{a}\|^2$.

$$X^T X \hat{a} = X^T \hat{y}, \quad \hat{a} = \underbrace{(X^T X)^{-1} X^T}_{X^+} \hat{y}$$

$X^+ =$ Moore–Penrose pseudoinverse ($= X^{-1}$ when X square). Idea: projects $\hat{y}$ onto the column space of X . **Trap:** $X^+ X = I$ holds; $XX^+ = I$ generally does NOT.

ML: classification

Features = inputs (x_1, x_2) ; plane

$a_0 + a_1x_1 + a_2x_2 = 0 \cap x_1x_2$ -plane = **decision**

boundary (class 1 vs 2). Logistic

$\sigma(z) = \frac{1}{1+e^{-z}} : \mathbb{R} \rightarrow [0, 1]$; classifier

$\sigma(a_0 + a_1x_1 + a_2x_2)$. **Worked mini-example:**

plane $-1 + x_1 + x_2 = 0$: $(0, 0) \rightarrow -1 < 0$ (class 1), $(1, 1) \rightarrow +1 > 0$ (class 2); σ gives $\approx 0.27, \approx 0.73$.

2 — Linearization

Goal: transform $(\hat{x}, \hat{y})$ so model is linear; fit; inverse-transform back. Logs turn $\times$ into $+$ and powers into $\times$. Steps: (1) change vars; (2) fit $\hat{y} = a_0 + a_1\hat{x}$; (3) recover A, B ; (4) check fit.

- $y = Ae^{Bx}$: $\hat{x} = x$, $\hat{y} = \ln y$; $a_0 = \ln A$, $a_1 = B \iff A = e^{a_0}$.
- $y = Ax^n$: $\hat{x} = \ln x$, $\hat{y} = \ln y$; $a_0 = \ln A$, $a_1 = n$.
- $y = A + \frac{B}{x}$: $\hat{x} = \frac{1}{x}$, $\hat{y} = y$; $a_0 = A$, $a_1 = B$.
- $y = \frac{1}{A+Bx}$: $\hat{x} = x$, $\hat{y} = \frac{1}{y}$; $a_0 = A$, $a_1 = B$.
- $y = \frac{Ax}{B+x}$: $\hat{x} = \frac{1}{x}$, $\hat{y} = \frac{1}{y}$; $a_0 = \frac{1}{A}$, $a_1 = \frac{B}{A} \iff A = \frac{1}{a_0}$, $B = \frac{a_1}{a_0}$.

Trap: not everything linearizes (e.g. $a_0e^{-x} \sin(a_1 + 2\pi x)$). **Model from graph:** straight $\rightarrow$ linear; explosive growth $\rightarrow Ae^{Bx}$; power curve $\rightarrow Ax^n$; saturating hyperbola $\rightarrow Ax/(B+x)$; reciprocal decay $\rightarrow A+B/x$.

3 — Grids + Riemann Sums

Partition $a = x_0 < \dots < x_n = b$, $h_i = x_i - x_{i-1}$, sample $x_i^* \in [x_{i-1}, x_i]$: $\int_a^b f \approx \sum f(x_i^*)h_i$.

Uniform: $h_i \equiv h = \frac{b-a}{n}$, $n+1$ points.

Non-uniform: varying h_i (adaptive: fine where f changes fast, coarse elsewhere). **Riemann integral** = $\lim_{\max h_j \rightarrow 0} \sum h_i f(x_i^*)$; f bounded, at most countably many jumps.

4 — Rectangle / Trapezoid / Simpson

Sample point: $x_i^* = \alpha x_{i-1} + (1-\alpha)x_i$: $\alpha = 1$ left, 0 right, $\frac{1}{2}$ mid. (Quiz7 $\alpha = 0.2$; Lec16 $x_{i-1} + h_i/3 \Rightarrow \alpha = \frac{2}{3}$, $+2h_i/3 \Rightarrow \alpha = \frac{1}{3}$). General non-uniform: left $\sum h_i f(x_{i-1})$, right $\sum h_i f(x_i)$, mid $\sum h_i f(\frac{x_{i-1}+x_i}{2})$.

$$\tilde{I}_{\text{Tr}} = \frac{1}{2} \sum_{i=1}^n (f(x_{i-1}) + f(x_i)) h_i$$

Uniform trapezoid: $\frac{h(f_0+f_n)}{2} + h \sum_{i=1}^{n-1} f_i$. **Trap idea:** one strip = rectangle hQ_1 + triangle $\frac{h}{2}(Q_2 - Q_1) = h(Q_1 + Q_2)/2$; uniform version averages neighbors (interior f_i counted twice $\rightarrow$ single sum). **Simpson** ($n = 2m$ even, uniform): fits parabola on each pair. Why $O(h^4)$? Parabola matches value+slope+curvature (trap: only value+slope).

$$I_{\text{parab}} = \frac{h}{3} \left\{ f_0 + f_n + 4 \sum_{\text{odd } i} f_i + 2 \sum_{\text{even } i < n} f_i \right\}$$

Accuracy estimates

- left/right rectangle $O(h)$; midpoint $O(h^2)$; trapezoid $O(h^2)$; Simpson $O(h^4)$.
- Read $O(h^p)$ as: halving h divides the error by 2^p (rect $\div 2$, mid/trap $\div 4$, Simpson $\div 16$).
- Left-rect global: $|E| \leq \frac{M_1(b-a)}{2} h$, $M_1 = \max |f'|$. Midpoint: $|E| \leq \frac{M_2(b-a)}{24} h^2$ (half the trapezoid constant $1/12$). (Taylor + symmetry kills f' term.)
- No elementary antiderivative ($\int e^{x^2} dx$, $\int \frac{dx}{\ln x}$): cross-check Simpson vs trapezoid.

5 — ODEs: IVP + Numeric Methods

$y' = f(x, y)$ (slope field = direction at each point). General solution $y(x, C)$ = whole family of curves; IVP (Cauchy) $+ y(x_0) = y_0$ picks the ONE curve through (x_0, y_0) (fix C by plugging IC; existence/uniqueness assumed). **Check-solution trap:** differentiate candidate AND plug into ODE — both must hold for ALL C . **Bridge (Lec19):** Euler comes from the finite difference $y' \approx \frac{y_{i+1} - y_i}{h}$: replace y' in $y' = f(x, y)$ and solve for y_{i+1} .

$$y_{i+1} = y_i + hf(x_i, y_i), \quad x_{i+1} = x_i + h \quad (\text{Euler})$$

$$y_{i+1} = y_i + hf(x_{i+\frac{1}{2}}, y_{i+\frac{1}{2}}), \quad x_{i+\frac{1}{2}} = x_i + \frac{h}{2}, \quad y_{i+\frac{1}{2}} = y_i + \frac{h}{2} f(x_i, y_i)$$

RK4: weights $(1, 2, 2, 1)/6$: $k_1 = f(x_n, y_n)$; $k_2 = f(x_n + \frac{h}{2}, y_n + \frac{h}{2}k_1)$; $k_3 = f(x_n + \frac{h}{2}, y_n + \frac{h}{2}k_2)$; $k_4 = f(x_n + h, y_n + hk_3)$.

$$y_{n+1} = y_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

Write-only schemes (Quiz9 Q4 + HW10 Q3):

for $y' = y + x$: $k_1 = y_i + x_i$, $k_2 = y_i + \frac{h}{2}k_1 + x_i + \frac{h}{2}$, $k_3 = y_i + \frac{h}{2}k_2 + x_i + \frac{h}{2}$, $k_4 = y_i + hk_3 + x_i + h$. For $y' = xy + 1/x$ ($h=1/12$): $k_1 = x_i y_i + 1/x_i$; $k_2 = (x_i + \frac{1}{24})(y_i + \frac{k_1}{24}) + 1/(x_i + \frac{1}{24})$; $k_3 = k_4 = (x_i + \frac{1}{12})(y_i + \frac{k_2}{12}) + 1/(x_i + \frac{1}{12})$. **Global errors:** Euler $O(h)$ (local $O(h^2) \times N \sim 1/h$); midpoint $O(h^2)$ (local $O(h^3)$, symmetric); RK4 $O(h^4)$ (4 evals/step).

6 — Linear Algebra

det = oriented area (2D) / volume (3D), up to sign; $\det \neq 0 \iff$ invertible. $\det(AB) = \det A \det B$.

$Ax = b$ unique $\iff \det A \neq 0 \iff n$ pivots.

Scalar product: $\bar{a} \cdot \bar{b} = a^x b^x + a^y b^y = |\bar{a}| |\bar{b}| \cos \alpha$ ($+a^z b^z$ in 3D); $|\bar{a}| = \sqrt{(a^x)^2 + (a^y)^2}$.

Basic ops: $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 5 \\ 6 \end{pmatrix} = \begin{pmatrix} 1 \cdot 5 + 2 \cdot 6 \\ 3 \cdot 5 + 4 \cdot 6 \end{pmatrix} = \begin{pmatrix} 17 \\ 39 \end{pmatrix}$ (row $\cdot$ column).

Projection: $\text{proj}_{\bar{b}} \bar{a} = \frac{\bar{a} \cdot \bar{b}}{|\bar{b}|} \frac{\bar{b}}{|\bar{b}|} = |\bar{a}| \cos \alpha$; basis:

$\text{proj}_{\bar{i}} \bar{a} = a^x$, $\text{proj}_{\bar{j}} \bar{a} = a^y$ (e.g. $(0, 1), (2, 0)$: $1, 2$).

Area: parallelogram $S = a^y b^x - a^x b^y = -\det$; area = $|S| = |\det|$. **Worked:** $u = (3, 1), v = (1, 2)$: $S = 1 \cdot 1 - 3 \cdot 2 = -5$, area 5. **Volume (3D):** $|\det|$ of the 3×3 ; **Worked:** $\text{diag}(1, 2, 3)$ volume = 6.

Non-diagonal: sheet's 3×3 A has $\det = -3 \Rightarrow$ volume 3.

Minor/cofactor: $\text{Minor}(a_{ij}) = \det$ deleting row i , col j ; $\text{cofac}(a_{ij}) = (-1)^{i+j} \text{Minor}(a_{ij})$. **Worked:** $\text{cofac}(a_{23}) = -(a_{11}a_{32} - a_{31}a_{12}) = a_{31}a_{12} - a_{11}a_{32}$. **Laplace:** $\det A = \sum_j a_{kj} \text{cofac}(a_{kj})$ (any row k).

Row swap flips sign. **Worked:**

$A = [[1, 2, 3], [4, 5, 6], [7, 8, 10]]$: cofactors 2, 2, $-3 \Rightarrow \det = 1(2) + 2(2) + 3(-3) = -3$.

Triangular: zeros above (lower) or below (upper) diagonal; $\det =$ product of diagonal. **Worked**

(5 $\times$ 5): $\text{diag}(1, 2, 3, 4, 5) \Rightarrow \det = 120 = 5!$.

LU: $A = LU$ (L unit-lower, U upper; set $\ell_{ii} = 1$).

Worked (2 $\times$ 2):

$[[1, 2], [3, 4]] = [[1, 0], [3, 1]][[1, 2], [0, -2]]$ ($u_{22} = 4 - 3 \cdot 2 = -2$). **Det:** $\det A = \det L \cdot \det U$;

here $1 \cdot (-2) = -2$ **Check** $= 1 \cdot 4 - 2 \cdot 3$.

Worked (3 $\times$ 3): $[[1, 2, 3], [4, 5, 6], [7, 8, 10]] = [[1, 0, 0], [4, 1, 0], [7, 2, 1]][[1, 2, 3], [0, -3, -6], [0, 0, 1]]$ ($\ell_{14}=4, u_{44}=-3, u_{55}=-6, \ell_{27}=7, \ell_{32}=2, u_{66}=1$); $\det = 1 \cdot (-3) \cdot 1 = -3$.

Solve $Ax = y$: $Ax = L(Ux) = y$; set $z = Ux$: solve $Lz = y$ (forward), then $Ux = z$ (backward).

Worked: A above, $y = [1, -1]^T$: $z_1 = 1$,

$3 + z_2 = -1 \Rightarrow z_2 = -4$; $-2z_2 = -4 \Rightarrow z_2 = 2$,

$x_1 + 4 = 1 \Rightarrow x_1 = -3$. $x = [-3, 2]^T$ **Check:**

$Ax = [1, -1]^T$.

7 — Midterm Recap (compact)

- Taylor:** $f(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k + R_n$, $|R_n| \leq \frac{M d^{n+1}}{(n+1)!}$; odd fn: only odd powers (deg 8 $\rightarrow x^7$), bound $\frac{1}{(2m+2)!}$.
- Maclaurin:** $e^x, \sin, \cos$ converge $\forall \mathbb{R}$; $\frac{1}{1-x} = \sum x^k$, $\ln(1+x)$, $\arctan$ for $|x| < 1$.
- Horner:** tableau at c gives $p(c)$ (last) + quotient q ; tableau on q gives $p'(c)$.
- Interpolation:** $\ell_i(x) = \prod_{j \neq i} \frac{x - x_j}{x_i - x_j}$, $p = \sum y_i \ell_i$; err = $\frac{f^{(n+1)}(\xi)}{(n+1)!} \prod (x - x_i)$ ($= 0$ at nodes).
- Vandermonde $\det = \prod_{i < j} (x_j - x_i) \neq 0$ (distinct nodes); DD: $f[x_i, x_{i+1}] = \frac{\Delta f}{\Delta x}$ (Newton form not required).

8 — Worked Mini-Examples

Coffee regression (Quiz 6)

Given $x : 0, 1, 2, 3$; $y : 3, 5, 8, 7$ ($n+1=4$ pts). **Step 1** — sums (add term by term):

$\sum x_i = 0+1+2+3 = 6$; $\sum y_i = 3+5+8+7 = 23$;

$\sum x_i^2 = 0+1+4+9 = 14$;

$\sum x_i y_i = 0+5+16+21 = 42$. **Step 2** — means

(divide by 4): $\bar{x} = 6/4 = 1.5$; $\bar{y} = 23/4 = 5.75$;

$\bar{x}^2 = 14/4 = 3.5$; $\bar{x}\bar{y} = 42/4 = 10.5$. **Step 3** —

coeffs (plug into boxed formula): $a_1 = \frac{10.5 - 1.5 \cdot 5.75}{3.5 - 1.5^2} = \frac{10.5 - 8.625}{3.5 - 2.25} = \frac{1.875}{1.25} = 1.5$; $a_0 = 5.75 - 1.5 \cdot 1.5$

$= 5.75 - 2.25 = 3.5$. $\hat{y} = 3.5 + 1.5x$. **Step 4** —

predict $x=4$: $\hat{y} = 3.5 + 1.5 \cdot 4 = 3.5 + 6 = 9.5$

(mark (4, 9.5) on the scatter plot with the line).

Plot recipe (Quiz6 Q2): scatter the 4 data points, draw the line through $(0, 3.5)$ and $(4, 9.5)$

— regression line need NOT pass through data points. **Step 5** — **Check fit:** $\hat{y} : 3.5, 5.0, 6.5, 8.0$

vs $y : 3, 5, 8, 7$; residuals $-0.5, 0, 1.5, -1.0$; squared error $S = 0.25 + 0 + 2.25 + 1 = 3.5$ (this S is what least squares minimized). **Hessian (Quiz6 Q4, coffee data)**: from Step 1 sums ($\sum x_i = 6$, $\sum x_i^2 = 14$): $S_{a_0 a_0} = 2 \cdot 4 = 8$, $S_{a_0 a_1} = 2 \cdot 6 = 12$, $S_{a_1 a_1} = 2 \cdot 14 = 28$; $\det H = 8 \cdot 28 - 12^2 = 224 - 144 = 80 > 0$ ($+8 > 0$) $\Rightarrow$ min.

Battery fit (Lec 14)

Given $t : 0, 10, 20, 30, 40, 50, 60$; charge $5, 10, 15, 25, 35, 40, 45$ ($n+1=7$). Sums: $\sum t_i = 0+10+20+30+40+50+60 = 210$; $\sum y_i = 5+10+15+25+35+40+45 = 175$; $\sum t_i^2 = 0+100+400+900+1600+2500+3600 = 9100$; $\sum t_i y_i = 0+100+300+750+1400+2000+2700 = 7250$. Means (divide by 7): $\bar{x} = 210/7 = 30$; $\bar{y} = 175/7 = 25$; $\overline{x^2} = 9100/7 = 1300$; $\overline{xy} = 7250/7 \approx 1035.71$. Coeffs: $a_1 = \frac{1035.71-30 \cdot 25}{1300-30^2} = \frac{1035.71-750}{1300-900} = \frac{285.71}{400} \approx 0.714$; $a_0 = 25 - 0.714285 \cdot 30 = 25 - 21.43 \approx 3.57$. $\boxed{\phi(x) = 0.714x + 3.57}$.

Check fit:

$\hat{y} : 3.57, 10.71, 17.85, 24.99, 32.13, 39.27, 46.41$ vs $y : 5, 10, 15, 25, 35, 40, 45$. **Predict $t=70$ (same battery data)**: $0.714 \cdot 70 + 3.57 = 53.55\%$ (extrapolation — flag as less reliable).

Linearization drills (HW09 Q4 + Lec15)

Given $x : 1, 2, 3, 5$, $y : 8, 4, 2, 1$; model $y = Ae^{Bx} \Rightarrow$ transform $\tilde{x} = x$, $\tilde{y} = \ln y$. **Step 0** — transform each y : $\ln 8 \approx 2.07944$, $\ln 4 \approx 1.38629$, $\ln 2 \approx 0.69315$, $\ln 1 = 0$. So $\tilde{x} : 1, 2, 3, 5$; $\tilde{y} : 2.07944, 1.38629, 0.69315, 0$. **Step 1** — means ($n+1=4$): $\bar{\tilde{x}} = (1+2+3+5)/4 = 11/4 = 2.75$; $\bar{\tilde{y}} = (2.07944+1.38629+0.69315+0)/4 = 4.15888/4 \approx 1.03972$; $\bar{\tilde{x}^2} = (1+4+9+25)/4 = 39/4 = 9.75$; $\overline{\tilde{x}\tilde{y}} = (1 \cdot 2.07944 + 2 \cdot 1.38629 + 3 \cdot 0.69315 + 5 \cdot 0)/4 = 6.93147/4 \approx 1.73287$. **Step 2** — fit: $a_1 = \frac{1.73287 - 2.75 \cdot 1.03972}{9.75 - 2.75^2} = \frac{1.73287 - 2.85923}{9.75 - 7.5625} = \frac{-1.12636}{2.1875} \approx -0.5149$; $a_0 = 1.03972 - (-0.5149) \cdot 2.75 = 1.03972 + 1.41598 \approx 2.4557$. **Step 3** — back-transform ($A = e^{a_0}$, $B = a_1$): $A = e^{2.4557} \approx 11.65$, $B \approx -0.515 < 0$. **Check**: $\hat{y} = 11.65e^{-0.515x}$ gives $7.0, 4.2, 2.5, 0.9$ vs $8, 4, 2, 1$ — reasonable.

Second model (Lec15):

$x : 1, 2, 3, 4$, $y : 0.45, 0.81, 1.05, 1.13$ for $y = \frac{A}{B+x}$: $\tilde{x} = 1/x$, $\tilde{y} = 1/y$ give $a_1 \approx 1.834$, $a_0 \approx 0.368$, so $A = 1/a_0 \approx 2.71$, $B = a_1/a_0 \approx 4.98$.

Third model — $y = A + B/x$:

$x : 1, 2, 4$, $y : 5, 3, 2$: $\tilde{x} = 1/x : 1, 0.5, 0.25$; means $\bar{\tilde{x}} = 0.58333$, $\bar{y} = 3.33333$, $\bar{\tilde{x}^2} = 0.4375$, $\overline{\tilde{x}y} = 2.33333$; $a_1 = B = \frac{2.33333 - 0.58333 \cdot 3.33333}{0.4375 - 0.58333^2} = \frac{0.38889}{0.09722} = 4.0$; $a_0 = A = 3.33333 - 4 \cdot 0.58333 = 1.0$. $\boxed{y = 1 + 4/x}$ (exact through all 3 pts here).

Trapezoid $\sin x$ (Quiz 7)

Given $[0, 3\pi/2]$, uniform $h = \pi/4 \Rightarrow n = (3\pi/2)/(\pi/4) = 6$ subintervals, 7 nodes $x_i = i\pi/4$. **Step 1** — evaluate $f(x) = \sin x$: $\sin 0 = 0$; $\sin \frac{\pi}{4} = \frac{\sqrt{2}}{2} \approx 0.70711$; $\sin \frac{\pi}{2} = 1$; $\sin \frac{3\pi}{4} = \frac{\sqrt{2}}{2}$; $\sin \pi = 0$; $\sin \frac{5\pi}{4} = -\frac{\sqrt{2}}{2}$; $\sin \frac{3\pi}{2} = -1$. **Step 2** — uniform trapezoid (endpoints half-weighted): $\tilde{I} = \frac{h}{2}(f_0 + f_6) + h \sum_{i=1}^5 f_i$ with $\sum_{i=1}^5 f_i = \frac{\sqrt{2}}{2} + 1 + \frac{\sqrt{2}}{2} + 0 - \frac{\sqrt{2}}{2} = 1 + \frac{\sqrt{2}}{2} \approx 1.70711$. **Step 3** — multiply out: $\tilde{I} = \frac{h}{2}(f_0 + f_6) + h \sum_{i=1}^5 f_i = \frac{\pi}{4} \left(\frac{1}{2}(-1) + 1.70711 \right) = \frac{\pi}{4} (1.20711) \approx 0.78540$. $1.20711 \approx 0.94806$. **Step 4** — exact $I = \int_0^{3\pi/2} \sin x \, dx = [-\cos x]_0^{3\pi/2} = -\cos \frac{3\pi}{2} + \cos 0 = 0 + 1 = 1$; err $= |1 - 0.94806| \approx 0.05194$. **Draw (Pract.6)**: mark $n+1$ nodes; plot $(x_i, f(x_i))$; join with segments (trapezoids); Simpson: one arc per pair through its 3 points.

Simpson $\int_2^4 \frac{dx}{\ln x}$ (Quiz 8)

Given $n = 8$ (even $\checkmark$), 9 points, $h = (4 - 2)/8 = 0.25$. Nodes $x_i = 2 + 0.25i$:

$2, 2.25, 2.5, 2.75, 3, 3.25, 3.5, 3.75, 4$. **Step 1** —

evaluate $f(x) = 1/\ln x$ at each node:

$f_0 = 1/\ln 2 \approx 1.44270$; $f_1 = 1/\ln 2.25 \approx 1.23315$; $f_2 = 1/\ln 2.5 \approx 1.09136$; $f_3 = 1/\ln 2.75 \approx 0.98853$; $f_4 = 1/\ln 3 \approx 0.91024$; $f_5 = 1/\ln 3.25 \approx 0.84842$; $f_6 = 1/\ln 3.5 \approx 0.79824$; $f_7 = 1/\ln 3.75 \approx 0.75657$; $f_8 = 1/\ln 4 \approx 0.72135$. **Step 2** — weight (ends $\times 1$, odd $i \times 4$, even $i \times 2$): $4 \sum_{\text{odd}} = 4(1.23315 + 0.98853 + 0.84842 + 0.75657) = 4(3.82667) = 15.30668$; $2 \sum_{\text{even}} = 2(1.09136 + 0.91024 + 0.79824) = 2(2.79984) = 5.59968$. **Step 3** — assemble: $I = \frac{0.25}{3} (1.44270 + 15.30668 + 5.59968 + 0.72135) = \frac{0.25}{3} (23.07041) \approx 0.83333 \cdot 23.07041 \approx 1.92253$.

_____ **Euler + midpoint** $y' = x - 4y$, $y(0) = -3/4$ (Quiz 9)

$h = 0.2$, $f(x, y) = x - 4y$. **Euler**:

$y_0 = 0 - 4(-0.75) = 3$;

$y_1 = -0.75 + 0.2(3) = -0.15$;

$f_1 = 0.2 - 4(-0.15) = 0.8$;

$y_2 = -0.15 + 0.2(0.8) = 0.01$;

$f_2 = 0.4 - 0.04 = 0.36$;

$y_3 = 0.01 + 0.2(0.36) = 0.082$. **Midpoint**:

half-step $(0.1, -0.75 + 0.1(3)) = (0.1, -0.45)$,

$f = 0.1 + 1.8 = 1.9$

$\rightarrow y_1 = -0.75 + 0.2(1.9) = -0.37$; half-step

$(0.3, -0.37 + 0.1(1.68)) = (0.3, -0.202)$,

$f_3 = 0.3 + 0.808 = 1.108$

$\rightarrow y_2 = -0.37 + 0.2(1.108) = -0.1484$. Exact

$y(0.4) \approx -0.101304$ (given);

$y = x/4 - 1/16 - \frac{11}{16}e^{-4x}$: Euler err

$= |-0.101304 - 0.01| \approx 0.11130$; midpoint err

$= |-0.101304 - (-0.1484)| \approx 0.04710$. **Quiz9 Q1**

— **IVP from general sol**: $y = -\frac{1}{4} + \frac{x}{2} + Ce^{-2x}$,

$(x_0, y_0) = (\frac{1}{2}, \frac{1}{4})$: $\frac{1}{4} = -\frac{1}{4} + \frac{1}{4} + Ce^{-1} \Rightarrow$

$C = e/4 \approx 0.6796$, $y^* = -\frac{1}{4} + \frac{x}{2} + \frac{e}{4}e^{-2x}$. **HW10**

Q2 pattern: $y' = \frac{y}{x} + 2x^2$, $(1, 4)$, general

$y = (x^2 + C)x$: $4 = (1 + C) \cdot 1 \Rightarrow C = 3$,

$y^* = (x^2 + 3)x$. Euler pattern: $x_1 = 1 + h$,

$y_1 = 4 + h(4/1 + 2) = 4 + 6h$ (since $f(1, 4) = 6$).

Midpoint pattern: $x_{1/2} = 1 + h/2$,

$y_{1/2} = 4 + h(h/2) \cdot 6 = 4 + 3h$. **Quiz9 Q2 — check**

trap (fail case): $y = Ce^x - 2 \cos 2x - 2 \sin 2x$ is

not a sol of $y' = y + 5 \sin 2x$:

$y' = Ce^x + 4 \sin 2x - 4 \cos 2x$, but

$y + 5 \sin 2x = Ce^x - 2 \cos 2x + 3 \sin 2x$; difference

$= \sin 2x - 2 \cos 2x \neq 0$.

Pass case (HW10 Q1):

$y = \frac{1}{2}(\cos x - \sin x) + Ce^x$ is a sol of $y' = y - \cos x$:

$y' = \frac{1}{2}(-\sin x - \cos x) + Ce^x = y - \cos x$.

Rule: differentiate candidate AND plug into ODE

— both must hold $\forall C$.

Rectangle lattice nodes (Quiz 7)

Given $x_k = \frac{3 \ln(1+k/5)}{\ln 2} - 1$, $k = 0..5$. Sample:

$x_1 = \frac{3 \ln 1.2}{\ln 2} - 1 \approx -0.21090$ (since

$3 \ln 1.2 \approx 0.54696$, $\ln 2 \approx 0.69315$);

$x_0 = \frac{3 \ln 1}{\ln 2} - 1 = -1$; $x_5 = \frac{3 \ln 2}{\ln 2} - 1 = 2$.

Full list: $-1.00000, -0.21090, 0.45628, 1.03422,$

$1.54399, 2.00000$. $x_i^* = 0.2x_{i-1} + 0.8x_i$: $-0.36872,$

$0.32285, 0.91863, 1.44204, 1.90880$.

Widths $h_i = x_i - x_{i-1}$:

$0.78910, 0.66718, 0.57794, 0.50977, 0.45601$; heights

$f(x_i^*)$ ($f = x^3 - 1.25x^2$):

$-0.22007, -0.09663, -0.27964, 0.39934, 2.40035$;

e.g. $R_1 = 0.78910 \cdot (-0.22007) \approx -0.17366$, total

$R = \sum h_i f(x_i^*) \approx 0.89843$.

Rect i has width $h_i = x_i - x_{i-1}$, height $f(x_i^*)$

($f = x^3 - 1.25x^2$). **How to draw (Pract.5)**: (1)

mark the $n+1$ nodes on the x -axis; (2) for each

subinterval compute x_i^* via the α -formula (any α);

(3) go up to the curve at $f(x_i^*)$ — that height is

the bar top; (4) draw the bar over $[x_{i-1}, x_i]$ (works

below the axis too: height negative $\Rightarrow$ signed area).

_____ **Accuracy table** $\int_0^1 x^2 dx$ (Lec17, $h = 0.25$)

Nodes $0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1$; $f(x) = x^2$ gives $0, \frac{1}{16}, \frac{1}{4}, \frac{9}{16}, 1$.

Left $= \frac{1}{64}$? **No**: $f_3 = (\frac{3}{4})^2 = \frac{9}{16}$, not $\frac{3}{4}$. Correct

left uses $f_0..f_3$: $= \frac{1}{4}(0 + \frac{1}{16} + \frac{1}{4} + \frac{9}{16}) = \frac{1}{4}(\frac{14}{16})$

$= \frac{7}{32} \approx 0.21875$, err

$= |0.33333 - 0.21875| \approx 0.11458$.

Mid: midpoints $\frac{1}{8}, \frac{3}{8}, \frac{5}{8}, \frac{7}{8}$, $f^* : \frac{9}{64}, \frac{9}{64}, \frac{25}{64}, \frac{49}{64}$;

$\tilde{I} = \frac{1}{4} \cdot \frac{8}{64} \approx 0.32813$ (err 0.00521).

Right $= \frac{1}{4}(\frac{1}{16} + \frac{1}{4} + \frac{9}{16} + 1) = \frac{15}{32} \approx 0.46875$ (err

0.13542).

Trap $= \frac{11}{32} \approx 0.34375$ (err 0.01042). **Derivation**

idea (Th.10): Taylor + integrate: local

$|E_i| \leq M_1 h_i^2/2$, sum $\Rightarrow |E| \leq \frac{M_1(b-a)}{2} h = O(h)$;

midpoint: odd term vanishes $\Rightarrow O(h^2)$.

Simpson $\int_1^2 e^{x^2} dx$ (Lec19, $n = 4$)

Given $h = (2 - 1)/4 = 0.25$, nodes $x_i = 1 + 0.25i$:

$1, \frac{5}{4}, \frac{3}{2}, \frac{7}{4}, 2$. **Step 1** — evaluate $f(x) = e^{x^2}$

(square x first, then exponentiate):

$f_0 = e^1 \approx 2.71828$;

$f_1 = e^{(5/4)^2} = e^{1.5625} \approx 4.77073$;

$f_2 = e^{(3/2)^2} = e^{2.25} \approx 9.48774$;

$f_3 = e^{(7/4)^2} = e^{3.0625} \approx 21.38094$;

$f_4 = e^4 \approx 54.59815$.

$I = \frac{0.25}{3}(2.71828 + 4(26.15167) + 2(9.48774) +$

$54.59815) \approx 15.0749$.

Trap check: two double-intervals

$T_1 = h(f_0 + f_2) = 0.25(2.71828 + 9.48774) =$

$0.25(12.20602) \approx 3.0515$,

$T_2 = h(f_2 + f_4) = 0.25(9.48774 + 54.59815) =$

$0.25(64.08589) \approx 16.0215$, total $\approx 19.0730 \rightarrow 19.07$

— same ballpark $\Rightarrow$ likely right.

Simpson sketch (Quiz8 Q1): $[0, 240]$, $h = 40$, 3

arcs (one per pair); mark nodes, each arc through

its 3 points.

Pseudoinverse $\rightarrow$ 1D formulas (HW09)

$X^T X \bar{a} = X^T \bar{y}$ with $X = [1 \ x_i]$ gives

$\left(\sum_{i=1}^{n+1} x_i \right) \begin{pmatrix} a_0 \\ a_1 \end{pmatrix} = \begin{pmatrix} \sum y_i \\ \sum x_i y_i \end{pmatrix}$; dividing by

$n+1$ and solving (Cramer) returns

$a_1 = \frac{\overline{xy} - \bar{x}\bar{y}}{\bar{x}^2 - \bar{x}^2}$, $a_0 = \bar{y} - a_1 \bar{x}$. Cramer in full:

$\Delta = (n+1) \left(\sum x_i^2 - (\sum x_i)^2 \right)$;

$a_0 = \frac{(\sum y_i)(\sum x_i^2) - (\sum x_i)(\sum x_i y_i)}{\Delta}$;

$a_1 = \frac{(n+1)(\sum x_i y_i) - (\sum x_i)(\sum y_i)}{\Delta}$; dividing top

and bottom by $(n+1)^2$ gives the averaged

formulas. **HW09 Q2 setup ($m=2, n=2$, 3 pts)**:

X is 3×3 (first col of 1's, then $x_1^{(i)}, x_2^{(i)}$ rows);

$X^+ = (X^T X)^{-1} X^T$ is 3×3 . **Q3**: $X^+ X = I$

always; $XX^+ = I$ only if X square nonsingular.

Euler $y' = y + x$ (Lec20, corrected)

$(x_0, y_0) = (0, 1)$, $h = 0.25$: $y_1 = 1 + 0.25(1) = 1.25$;

$y_2 = 1.25 + 0.25(1.5) = 1.625$;

$y_3 = 1.625 + 0.25(2.125) = 2.15625$. (Compiled

$1.5625/2.078125$ used $f_1 = 1.25$ instead of

$0.25 + 1.25 = 1.5$ — knock-on slip.)

Exact $y^* = 2e^x - x - 1$: $y(0.5) \approx 1.79744$,

$y(0.75) \approx 2.48400$ — Euler visibly below.

Midpoint on same ODE:

$(0.125, 1.125) \rightarrow y_1 = 1 + 0.25(1.25) = 1.3125$ vs

exact 1.31805 — much closer.

Separate ODEs (Lec20):

$y' = x/y \Rightarrow y \, dy = x \, dx \Rightarrow y^2 = x^2 + C_1$. Circle

IVP $y' = -x/y$, $(2, 2)$: $y^2/2 + x^2/2 = C = 4$

(radius 2).

IVP slip (same ODE, HW pattern):

$y' = y + x$, $(1, 4)$: $4 = Ce - 2 \Rightarrow C = 6/e \approx 2.207$

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